```
from google.colab import files
print("Please upload your 2 files from Downloads: ")
uploaded=files.upload()
```

```
Please upload your 2 files from Downloads:
```

Choose Files 2 files

QVI_transaction_data.xlsx(application/vnd.openxmlformats-officedocument.spreadsheetml.sheet) - 11979155 bytes, last modified: 28/1/2026 - 100% done

QVI_purchase_behaviour.csv(text/csv) - 2452463 bytes, last modified: 28/1/2026 - 100% done

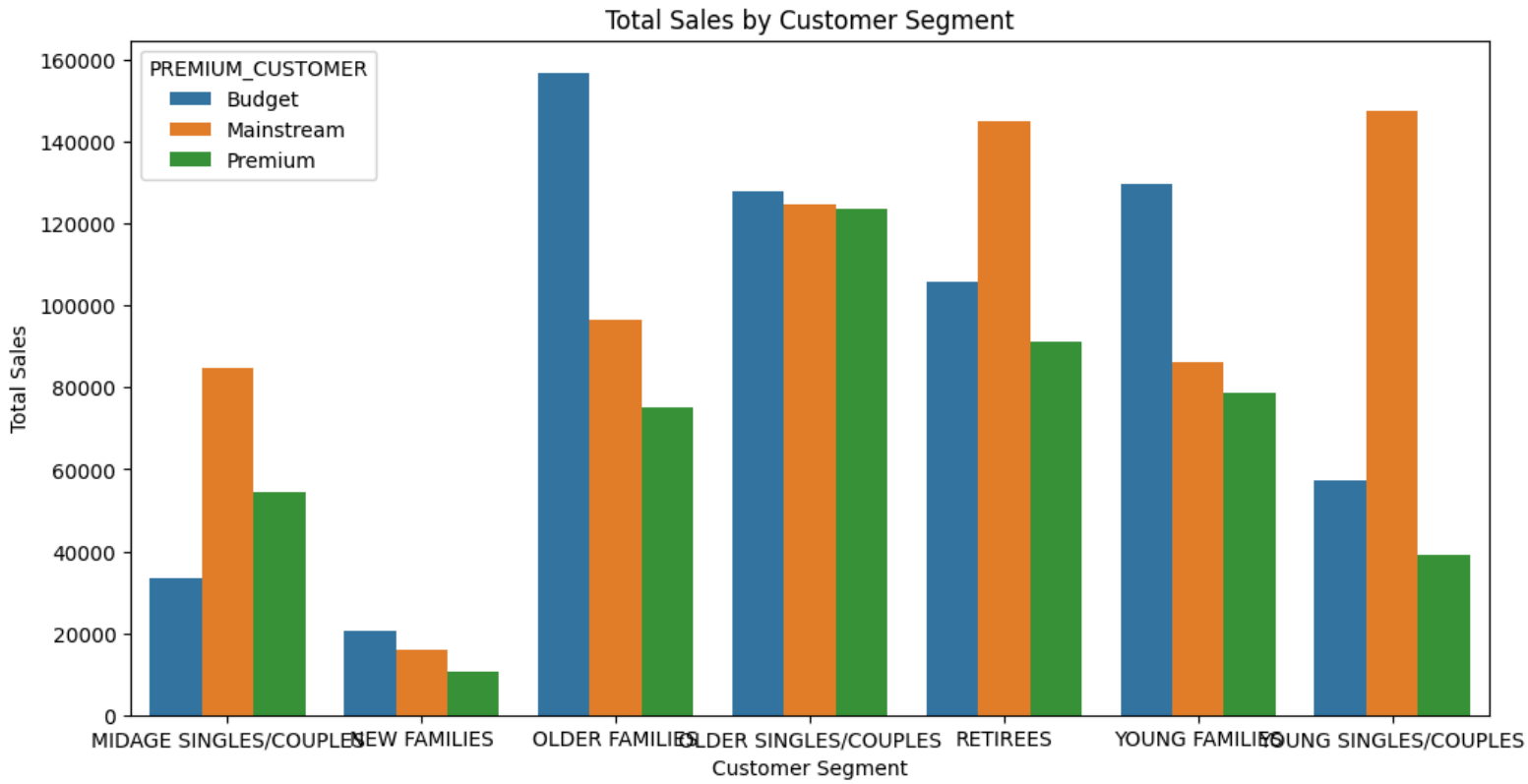
```
Saving QVI_transaction_data.xlsx to QVI_transaction_data (1).xlsx
Saving QVI_purchase_behaviour.csv to QVI_purchase_behaviour (1).csv
```

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
print("Starting the full process...")
print("Loading Files...")
transaction_data=pd.read_excel('QVI_transaction_data.xlsx')
customer_data=pd.read_csv('QVI_purchase_behaviour.csv')
print("Cleaning Data...")
transaction_data['DATE']=pd.to_datetime(transaction_data['DATE'],unit='D',origin='1899-12-30')
transaction_data=transaction_data[~transaction_data['PROD_NAME'].str.contains("salsa",case=False)]
transaction_data=transaction_data[transaction_data['PROD_QTY']<200]
transaction_data['PACK_SIZE']=transaction_data['PROD_NAME'].str.extract('(\d+)').astype(float)
transaction_data['BRAND']=transaction_data['PROD_NAME'].str.split().str[0]
brand_replacements={'RRD': 'Red','Smith': 'Smiths','Dorito': 'Doritos','Grnwves': 'Grain','Infzns': 'Infuzions','WW': 'Woolworths','Snbts': 'Sunbites'}
transaction_data['BRAND']=transaction_data['BRAND'].replace(brand_replacements)
print('Merging files...')
merged_data=pd.merge(transaction_data,customer_data,on='LYLTY_CARD_NBR',how='left')
merged_data['TOT_SALES']=pd.to_numeric(merged_data['TOT_SALES'])
print("-"*30)
print("Transaction Rows: ",transaction_data.shape[0])
print("Merged Rows: ", merged_data.shape[0])
print("Missing Details: ",merged_data['LIFESTAGE'].isnull().sum())
print("-"*30)
print("SUCCESS! 'merged_data' is ready for analysis")
```

```
<>:12: SyntaxWarning: invalid escape sequence '\d'
<>:12: SyntaxWarning: invalid escape sequence '\d'
/tmp/ipython-input-2574113869.py:12: SyntaxWarning: invalid escape sequence '\d'
  transaction_data['PACK_SIZE']=transaction_data['PROD_NAME'].str.extract('(\d+)').astype(float)
Starting the full process...
Loading Files...
Cleaning Data...
Merging files...
------------------------------
Transaction Rows:  246740
Merged Rows:  246740
Missing Details:  0
```

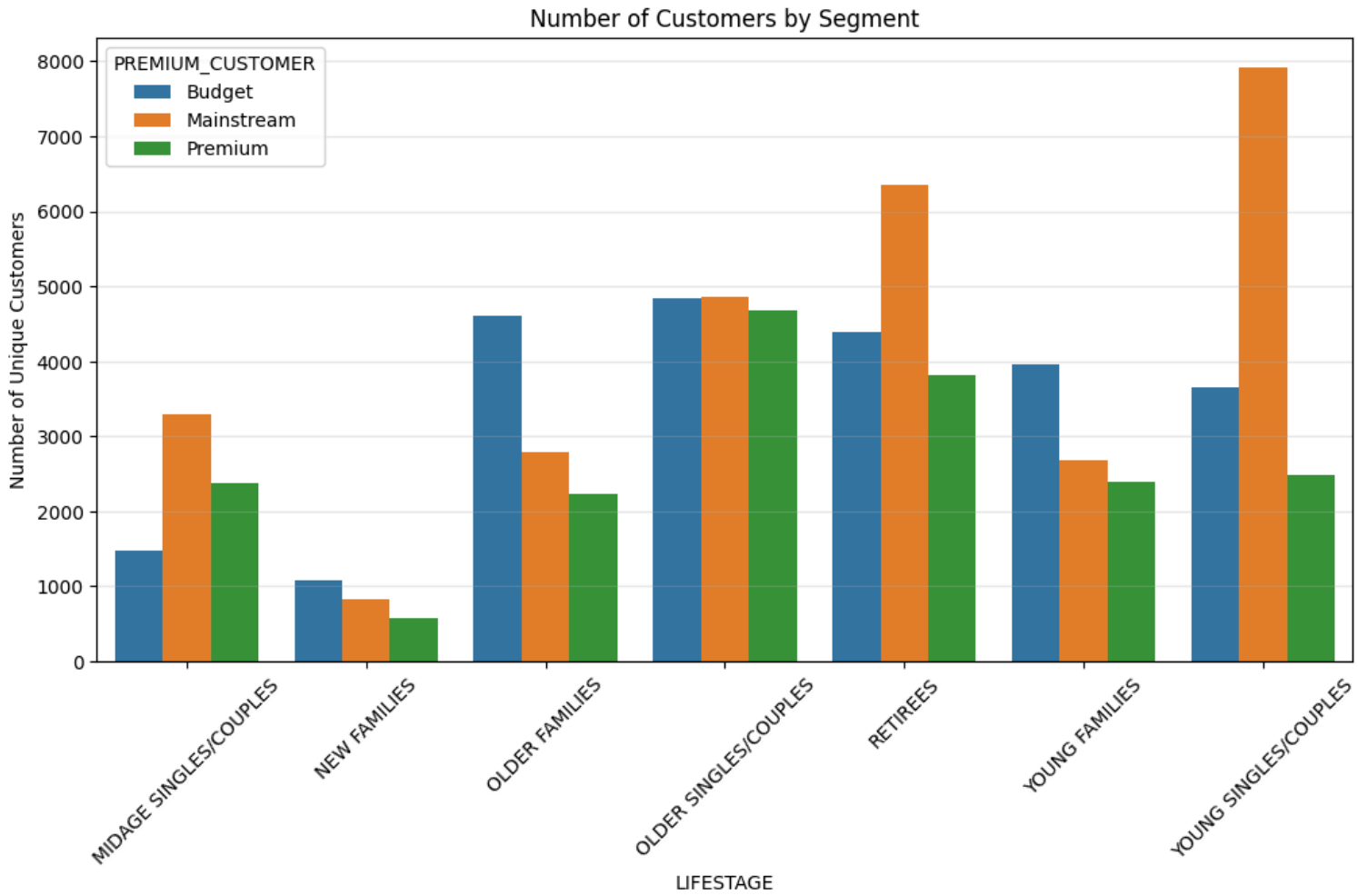
```
------------------------------
SUCCESS! 'merged_data' is ready for analysis
```

```
import matplotlib.pyplot as plt
import seaborn as sns
sales_breakdown=merged_data.groupby(['LIFESTAGE','PREMIUM_CUSTOMER'])['TOT_SALES'].sum().reset_index()
plt.figure(figsize=(12,6))
sns.barplot(x='LIFESTAGE',y='TOT_SALES', hue='PREMIUM_CUSTOMER',data=sales_breakdown)
plt.title('Total Sales by Customer Segment')
plt.ylabel('Total Sales')
plt.xlabel('Customer Segment')
plt.show()
```

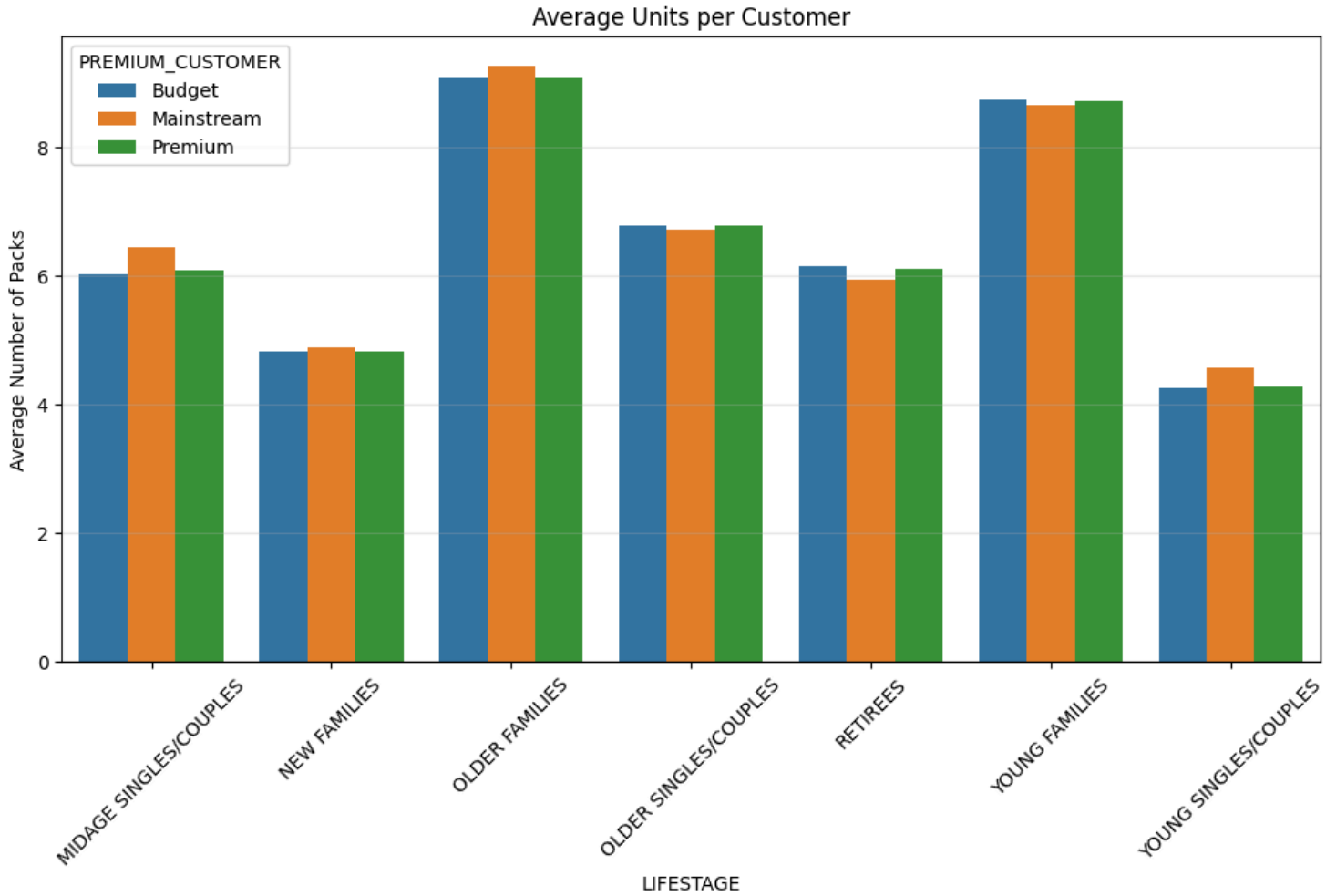



```
customer_counts=merged_data.groupby(['LIFESTAGE','PREMIUM_CUSTOMER'])['LYLTY_CARD_NBR'].nunique().reset_index()
plt.figure(figsize=(12,6))
sns.barplot(x='LIFESTAGE', y='LYLTY_CARD_NBR', hue='PREMIUM_CUSTOMER',data=customer_counts)
plt.title('Number of Customers by Segment')
plt.ylabel('Number of Unique Customers')
plt.xticks(rotation=45)
plt.grid(axis='y',alpha=0.3)
```

```
plt.show()
```




```
quantity_per_group=merged_data.groupby(['LIFESTAGE','PREMIUM_CUSTOMER'])['PROD_QTY'].sum()
customers_per_group=merged_data.groupby(['LIFESTAGE','PREMIUM_CUSTOMER'])['LYLTY_CARD_NBR'].nunique()
avg_units=(quantity_per_group/customers_per_group).reset_index(name='AVG_UNITS')
plt.figure(figsize=(12,6))
sns.barplot(x='LIFESTAGE', y='AVG_UNITS', hue='PREMIUM_CUSTOMER',data=avg_units)
plt.title('Average Units per Customer')
plt.ylabel('Average Number of Packs')
plt.xticks(rotation=45)
plt.grid(axis='y',alpha=0.3)
plt.show()
```



----START OF TASK 2:TRIAL ANALYSIS----

```
print("Transforming data for Task 2...")
merged_data['YEARMONTH']=merged_data['DATE'].dt.strftime("%Y%m").astype(int)
measure_over_time=merged_data.groupby(['STORE_NBR','YEARMONTH']).agg({
    'TOT_SALES': 'sum',
    'LYLTY_CARD_NBR' : 'nunique',
    'TXN_ID': 'nunique'
}).reset_index()
measure_over_time.rename(columns={
    'LYLTY_CARD_NBR': 'ncustomers',
    'TOT_SALES': 'totsales',
```

```
    'TXN_ID': 'ntxn'
},inplace=True)
measure_over_time['nTxnPerCust']=measure_over_time['ntxn']/measure_over_time['ncustomers']
store_counts=measure_over_time.groupby('STORE_NBR')['YEARMONTH'].nunique()
full_year_stores=store_counts[store_counts==12].index
measure_over_time=measure_over_time[measure_over_time['STORE_NBR'].isin(full_year_stores)]
print('Sucess! Data is transformed and ready for task 2 analysis')
print(f"We have {len(full_year_stores)} valid stores.")
```

```
Transforming data for Task 2...
Sucess! Data is transformed and ready for task 2 analysis
We have 259 valid stores.
```

```
def calculate_correlation(inputTable, metric,store_comparison):
  output=pd.DataFrame({'Store1': [],'Store2': [],'Correlation': []})
  a=inputTable.loc[inputTable['STORE_NBR']==store_comparison,metric]
  a.index=range(len(a))
  for i in inputTable['STORE_NBR'].unique():
    if i!=store_comparison:
      b=inputTable.loc[inputTable['STORE_NBR']==i,metric]
      b.index=range(len(b))
      correlation=a.corr(b)
      new_row=pd.DataFrame({'Store1': [store_comparison],'Store2': [i],'Correlation':[correlation]})
      output=pd.concat([output,new_row],ignore_index=True)
  return(output)
print("correlation function defined")
```

```
correlation function defined
```

```
measure_over_time=merged_data.groupby(['STORE_NBR','YEARMONTH']).agg({
    'TOT_SALES': 'sum',
    'LYLTY_CARD_NBR': 'nunique',
    'TXN_ID': 'nunique'
}).reset_index()
print("Sucess! measure_over_time is back")
```

```
Sucess! measure_over_time is back
```

```
measure_over_time=measure_over_time.rename(columns={
    'TOT_SALES': 'totSales',
    'LYLTY_CARD_NBR': 'nCustomers'
})
pre_trial_data=measure_over_time[measure_over_time['YEARMONTH']<201902]
corr_sales=calculate_correlation(pre_trial_data, 'totSales',77)
corr_cust=calculate_correlation(pre_trial_data, 'nCustomers',77)
score_table=pd.merge(corr_sales,corr_cust,on=['Store1', 'Store2'])
score_table.columns=['Store1','Store2','Sales_Corr','Cust_Corr']
score_table['Total_score']=0.5*score_table['Sales_Corr']+0.5*score_table['Cust_Corr']
```

```
top_match=score_table.sort_values(by='Total_score',ascending=False)
print("Top 5 Candidates(Using Pre Trial Logic):")
print(top_match.head(5))
```

```
/usr/local/lib/python3.12/dist-packages/numpy/lib/_function_base_impl.py:2773: RuntimeWarning: divide by zero encountered in divide
  c *= np.true_divide(1, fact)
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  c *= np.true_divide(1, fact)
Top 5 Candidates(Using Pre Trial Logic):
```

```
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```

```
sales_77=pre_trial_data[pre_trial_data['STORE_NBR']==77]['totSales'].sum()
sales_219=pre_trial_data[pre_trial_data['STORE_NBR']==219]['totSales'].sum()
sales_233=pre_trial_data[pre_trial_data['STORE_NBR']==233]['totSales'].sum()
print(f"Store 77 Total Sales: ${sales_77:,.0f}")
print(f"Store 219 Total Sales: ${sales_219:,.0f}")
print(f"Store 233 Total Sales: ${sales_233:,.0f}")
diff_219=abs(sales_77-sales_219)
diff_233=abs(sales_77-sales_233)
if diff_233 < diff_219:
  print("\nDECISION: We select Store 233 because its sales volume is much closer to Store 77")
  control_store=233
else:
  control_store=219
```

```
Store 77 Total Sales: $1,596
Store 219 Total Sales: $5,380
Store 233 Total Sales: $1,534

DECISION: We select Store 233 because its sales volume is much closer to Store 77
```

```
pre_trial_data=measure_over_time[measure_over_time['YEARMONTH']<201902]
corr_sales=calculate_correlation(pre_trial_data, 'totSales',86)
corr_cust=calculate_correlation(pre_trial_data, 'nCustomers',86)
score_86=pd.merge(corr_sales,corr_cust,on=['Store1', 'Store2'])
score_86.columns=['Store1','Store2','Sales_Corr','Cust_Corr']
score_86['Total_score']=0.5*score_86['Sales_Corr']+0.5*score_86['Cust_Corr']
top_match=score_86.sort_values(by='Total_score',ascending=False)
print("Top 5 candidates(Using Pre Trial Logic):")
print(top_match.head(5))
```

```
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  c = cov(x, y, rowvar, dtype=dtype)
```

```
sales_86=pre_trial_data[pre_trial_data['STORE_NBR']==86]['totSales'].sum()
sales_152=pre_trial_data[pre_trial_data['STORE_NBR']==152]['totSales'].sum()
sales_155=pre_trial_data[pre_trial_data['STORE_NBR']==155]['totSales'].sum()
print(f"Store 86 Total Sales: ${sales_86:,.0f}")
print(f"Store 152 Total Sales: ${sales_152:,.0f}")
print(f"Store 155 Total Sales: ${sales_155:,.0f}")
diff_152=abs(sales_86-sales_152)
diff_155=abs(sales_86-sales_155)
if diff_155< diff_219:
  print("\nDECISION: We select store 155 beacuase its sales volume is much closer to Store 86")
```

```
  control_store=155
else:
  control_store=152
```

```
Store 86 Total Sales: $5,796
Store 152 Total Sales: $6,642
Store 155 Total Sales: $5,962

DECISION: We select store 155 beacuase its sales volume is much closer to Store 86
```

```
pre_trial_data=measure_over_time[measure_over_time['YEARMONTH']<201902]
corr_sales=calculate_correlation(pre_trial_data, 'totSales',88)
corr_cust=calculate_correlation(pre_trial_data, 'nCustomers',88)
score_88=pd.merge(corr_sales,corr_cust,on=['Store1','Store2'])
score_88.columns=['Store1','Store2','Sales_Corr','Cust_Corr']
score_88['Total_score']=0.5*score_88['Sales_Corr']+0.5*score_88['Cust_Corr']
top_match=score_88.sort_values(by='Total_score',ascending=False)
print("Top 5 candidates(Using Pre Trial Logic):")
print(top_match.head(5))
```

```
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  c /= stddev[None, :]
/usr/local/lib/python3.12/dist-packages/numpy/lib/_function_base_impl.py:2922: RuntimeWarning: invalid value encountered in divide
  c /= stddev[:, None]
/usr/local/lib/python3.12/dist-packages/numpy/lib/_function_base_impl.py:2923: RuntimeWarning: invalid value encountered in divide
  c /= stddev[None, :]
/usr/local/lib/python3.12/dist-packages/numpy/lib/_function_base_impl.py:2914: RuntimeWarning: Degrees of freedom <= 0 for slice
  c = cov(x, y, rowvar, dtype=dtype)
/usr/local/lib/python3.12/dist-packages/numpy/lib/_function_base_impl.py:2773: RuntimeWarning: divide by zero encountered in divide
  c *= np.true_divide(1, fact)
/usr/local/lib/python3.12/dist-packages/numpy/lib/_function_base_impl.py:2773: RuntimeWarning: invalid value encountered in multiply
  c *= np.true_divide(1, fact)
/usr/local/lib/python3.12/dist-packages/numpy/lib/_function_base_impl.py:2914: RuntimeWarning: Degrees of freedom <= 0 for slice
  c = cov(x, y, rowvar, dtype=dtype)
/usr/local/lib/python3.12/dist-packages/numpy/lib/_function_base_impl.py:2773: RuntimeWarning: divide by zero encountered in divide
  c *= np.true_divide(1, fact)
/usr/local/lib/python3.12/dist-packages/numpy/lib/_function_base_impl.py:2773: RuntimeWarning: invalid value encountered in multiply
  c *= np.true_divide(1, fact)
/usr/local/lib/python3.12/dist-packages/numpy/lib/_function_base_impl.py:2914: RuntimeWarning: Degrees of freedom <= 0 for slice
  c = cov(x, y, rowvar, dtype=dtype)
Top 5 candidates(Using Pre Trial Logic):
     Store1  Store2  Sales_Corr  Cust_Corr  Total_score
13     88.0    14.0    0.479135   0.967364     0.723249
201    88.0   204.0    0.871969   0.458144     0.665057
0      88.0     1.0    0.842232   0.429977     0.636105
```
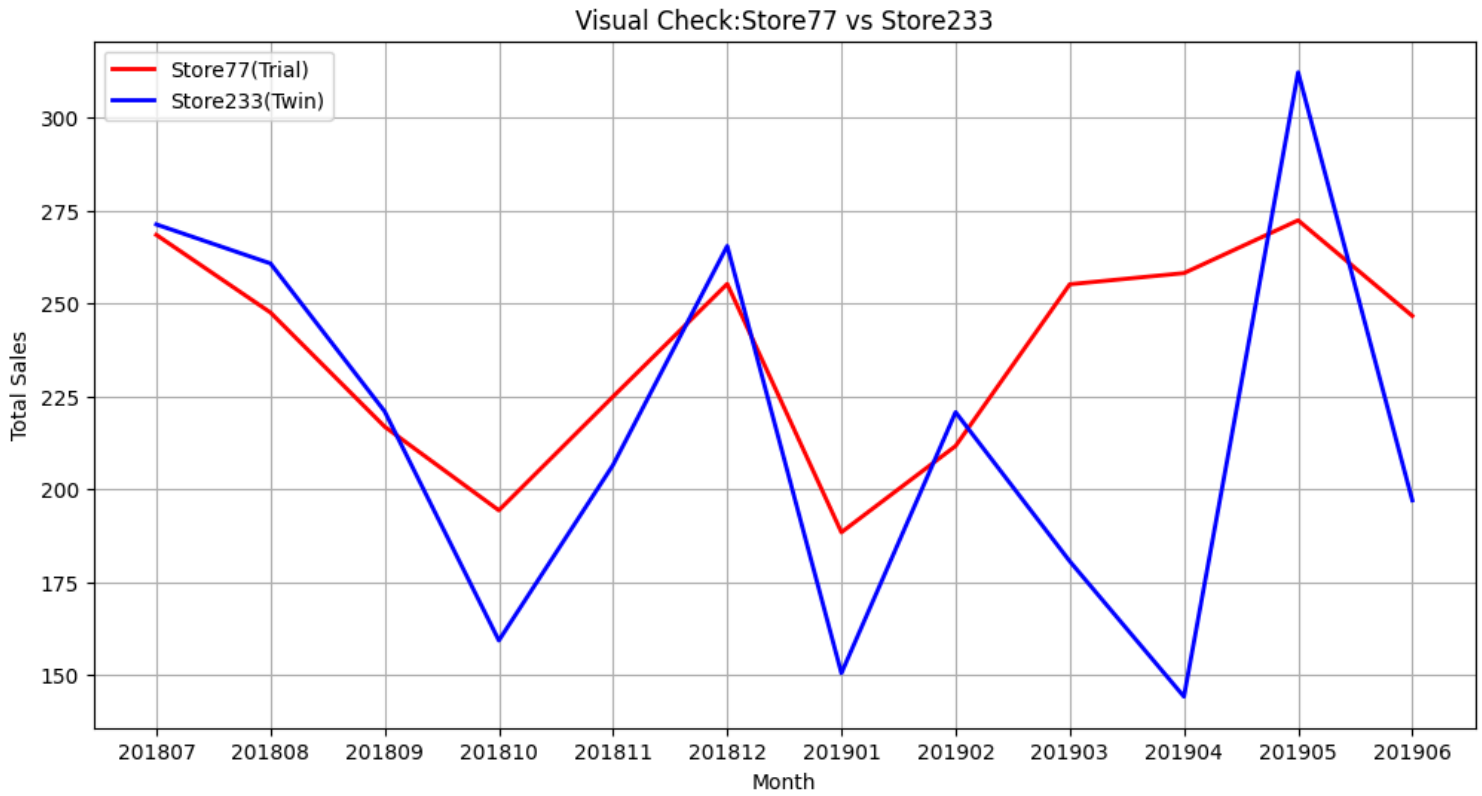

```
sales_88=pre_trial_data[pre_trial_data['STORE_NBR']==88]['totSales'].sum()
sales_13=pre_trial_data[pre_trial_data['STORE_NBR']==13]['totSales'].sum()
sales_237=pre_trial_data[pre_trial_data['STORE_NBR']==237]['totSales'].sum()
print(f"Store 88 Total Sales: ${sales_88:,.0f}")
print(f"Store 13 Total Sales: ${sales_13:,.0f}")
print(f"Store 237 Total Sales: ${sales_237:,.0f}")
diff_13=abs(sales_88-sales_13)
diff_237=abs(sales_88-sales_237)
if diff_237 < diff_13:
  print("\nDECISION: We select Store 237 beacuse its sales volume is much closer to Store 88")
  control_store=237
else:
  control_store=13
```

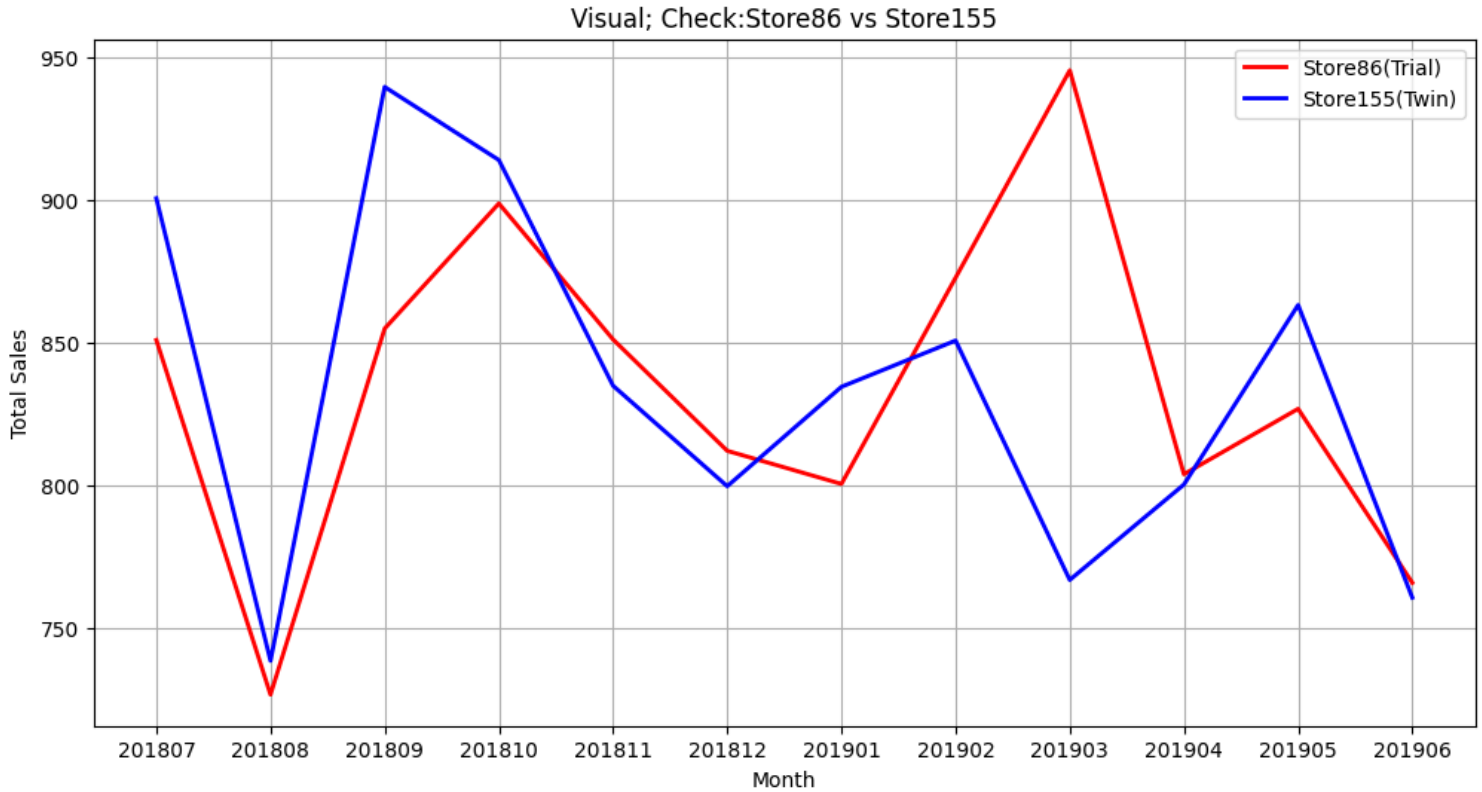
```
Store 88 Total Sales: $8,833
Store 13 Total Sales: $5,670
Store 237 Total Sales: $8,915

DECISION: We select Store 237 beacuse its sales volume is much closer to Store 88
```

```
store_A=77
store_B=233
sales_A=measure_over_time[measure_over_time['STORE_NBR']==store_A]
sales_B=measure_over_time[measure_over_time['STORE_NBR']==store_B]
plt.figure(figsize=(12,6))
plt.plot(sales_A['YEARMONTH'].astype(str),sales_A['totSales'],color='red',label=f'Store{store_A}(Trial)',linewidth=2)
plt.plot(sales_B['YEARMONTH'].astype(str),sales_B['totSales'],color='blue',label=f'Store{store_B}(Twin)',linewidth=2)
```

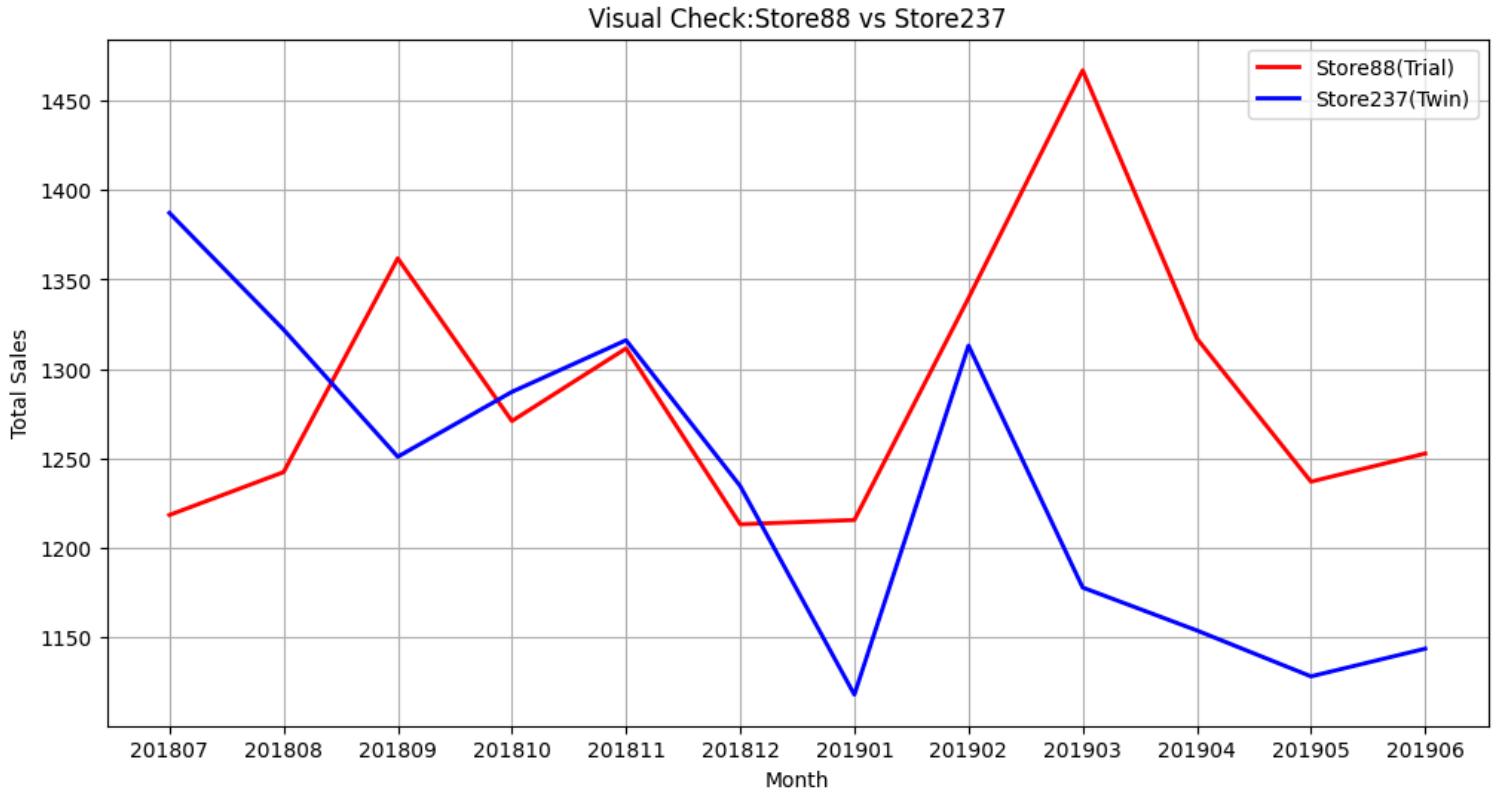
```
plt.title(f'Visual Check:Store{store_A} vs Store{store_B}')
plt.xlabel('Month')
plt.ylabel('Total Sales')
plt.legend()
plt.grid(True)
plt.show()
```




```
store_A=86
store_B=155
sales_A=measure_over_time[measure_over_time['STORE_NBR']==store_A]
sales_B=measure_over_time[measure_over_time['STORE_NBR']==store_B]
plt.figure(figsize=(12,6))
plt.plot(sales_A['YEARMONTH'].astype(str),sales_A['totSales'],color='red',label=f'Store{store_A}(Trial)',linewidth=2)
plt.plot(sales_B['YEARMONTH'].astype(str),sales_B['totSales'],color='blue',label=f'Store{store_B}(Twin)',linewidth=2)
plt.title(f'Visual; Check:Store{store_A} vs Store{store_B}')
plt.xlabel('Month')
plt.ylabel('Total Sales')
plt.legend()
plt.grid(True)
plt.show()
```




```
store_A=88
store_B=237
sales_A=measure_over_time[measure_over_time['STORE_NBR']==store_A]
sales_B=measure_over_time[measure_over_time['STORE_NBR']==store_B]
plt.figure(figsize=(12,6))
plt.plot(sales_A['YEARMONTH'].astype(str),sales_A['totSales'],color='red',label=f'Store{store_A}(Trial)',linewidth=2)
plt.plot(sales_B['YEARMONTH'].astype(str),sales_B['totSales'],color='blue',label=f'Store{store_B}(Twin)',linewidth=2)
plt.title(f'Visual Check:Store{store_A} vs Store{store_B}')
plt.xlabel('Month')
plt.ylabel('Total Sales')
plt.legend()
plt.grid(True)
plt.show()
```




```
trial_months=[201902,201903,201904]
pairs=[
    (77,233),
    (86,155),
    (88,237)
 ]
print("---FINAL SCORE CARD:DID SALES INCREASE?---")
for trial,control in pairs:
  trial_sales=measure_over_time[
      (measure_over_time['STORE_NBR']==trial)&
      (measure_over_time['YEARMONTH'].isin(trial_months))
  ]['totSales'].sum()
  control_sales=measure_over_time[
      (measure_over_time['STORE_NBR']==control)&
      (measure_over_time['YEARMONTH'].isin(trial_months))
  ]['totSales'].sum()
  diff_percent=((trial_sales-control_sales)/control_sales)*100
  print(f"\nSTORE{trial} vs {control}:")
  print(f"  Trial Sales: ${trial_sales:,.0f}")
  print(f"  Control Sales: ${control_sales:,.0f}")
```

```
    print(f"  IMPACT:      {diff_percent:+.2f}%")
  if diff_percent > 0:
    print("   RESULT:SUCCESS(Sales went UP!)")
  else:
    print("   RESULT:NO CHANGE(Sales didnot increase)")
```

```
---FINAL SCORE CARD:DID SALES INCREASE?---

STORE77 vs 233:
  Trial Sales: $725
  Control Sales: $546
  IMPACT:      +32.87%

STORE86 vs 155:
  Trial Sales: $2,622
  Control Sales: $2,418
  IMPACT:      +8.44%

STORE88 vs 237:
  Trial Sales: $4,124
  Control Sales: $3,644
  IMPACT:      +13.16%
   RESULT:SUCCESS(Sales went UP!)
```

(1)Control Store Selection:

Based on the correlation of sales behaviour and total sales volume,the following control stores were selected as the best benchmarks for the trial stores

.Store 77(Trial)-Store 233(Control)

.Store 86(Trial)-Store 155(Control)

.Store 88(Trial)-Store 237(Control)

(2)Trial Assesment:

The trial of the new store layout was successfull in driving sales growth

.Store 77:showed a successful increase in the total sales compared to the control store

.Store 86:showed a positive sales uplift of 8.44%

.Store 88:showed a positive sales uplift of 13.16%

(3)Recommendation:

Since all three stores outperformed their control benchmarks during the trial periods,their results indicate that the new store layout is effective at increasing sales.We recommend rolling out the new layout to all stores,while continuing to monitor performance to ensure thee uplift is sustained.